

# Finite-Sample Properties of Dynamic Panel Data Estimators: A Monte Carlo Investigation of the Nickell Bias

**Samuel Antonio Canedo Perez**

Course: Special Topics in Panel Data Econometrics

Wang Yanan Institute for Studies in Economics (WISE)

Xiamen University

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## **Abstract**

Static panel models assume that the current value of a variable is not influenced by its previous values. Nevertheless, in practice, in several fields, the present is highly influenced by past performance. This means that the dynamic model becomes relevant. The first econometric difficulty that arises is that using dynamic models in the framework of static panels as FE will produce extremely biased estimations. This means that the dynamic model becomes relevant. The term paper will provide a theoretical explanation for this problem, which is called Nickell bias. Using Monte Carlo simulations, we will provide empirical evidence for the theoretical asymptotic properties of Pooled OLS and Within-Group estimation. Moreover, we will evaluate the finite-sample properties of the Anderson-Hsiao IV estimator, demonstrating its asymptotic consistency as  $N$  grows, as well as its relative inefficiency in very short panels.

# Contents

|          |                                                                            |           |
|----------|----------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                        | <b>1</b>  |
| <b>2</b> | <b>Theoretical Framework and Methodology</b>                               | <b>2</b>  |
| 2.1      | The Dynamic Panel Data Model . . . . .                                     | 2         |
| 2.2      | The Inconsistency of Pooled OLS . . . . .                                  | 3         |
| 2.3      | The Within-Group Transformation and Nickell Bias . . . . .                 | 3         |
| 2.4      | The Anderson-Hsiao Instrumental Variable Approach . . . . .                | 4         |
| <b>3</b> | <b>Monte Carlo Simulation Design</b>                                       | <b>5</b>  |
| 3.1      | Data Generating Process (DGP) . . . . .                                    | 5         |
| 3.2      | Parameter Space and Estimation . . . . .                                   | 6         |
| <b>4</b> | <b>Research Content and Results</b>                                        | <b>6</b>  |
| 4.1      | The Upward Bias of Pooled OLS (The Upper Bound) . . . . .                  | 6         |
| 4.2      | The Nickell Bias Decay in Fixed Effects (The Lower Bound) . . . . .        | 7         |
| 4.3      | Finite-Sample Vulnerability of the Anderson-Hsiao (IV) Estimator . . . . . | 7         |
| <b>5</b> | <b>Conclusion</b>                                                          | <b>10</b> |
| <b>A</b> | <b>Supplementary Material: Python Source Code</b>                          | <b>12</b> |
| A.1      | Data Generating Process ( <code>dgp.py</code> ) . . . . .                  | 12        |
| A.2      | Estimators ( <code>estimators.py</code> ) . . . . .                        | 13        |
| A.3      | Visualization ( <code>graphs.py</code> ) . . . . .                         | 15        |
| A.4      | Monte Carlo Engine ( <code>monte_carlo.py</code> ) . . . . .               | 17        |

# 1 Introduction

The estimation of panel data models has emerged as an essential component of contemporary econometrics, providing distinct advantages relative to purely cross-sectional or time-series data. With the ability to observe multiple cross-sectional entities (individuals, firms, or geographical locations) over time, panel data allows the accounting for time-invariant unobservables. Yet, in many vital economic applications, it is unreasonable to rely on static relationships. Specifically, for example, in optimizing the design of robust global supply chain systems, especially in fast-developing industries such as the one of electric vehicles (EV), inventory holdings, manufacturing capacity, or procurement strategies in a certain time period are closely dependent on the choices that were made in previous time periods. This fact necessitates the consideration of dynamics of relationships, which involves dynamic panel data models.

While dynamic models are able to model the autoregressive property of economic processes, such as supply chain resilience, they give rise to serious econometric problems. The use of the lagged dependent variable breaks the assumption of strict exogeneity necessary for the consistency of the conventional panel estimators. Estimation of the dynamic panel data model using pooled OLS results in the absence of control of the unobserved individual heterogeneity and hence produces an upward-biased estimate. On the other hand, if we use the standard within-group transformation, then we are able to eliminate the unobserved heterogeneity, but the lagged dependent variable becomes negatively correlated with the error term after transformation. This result, as established by Nickell (1981), leads to the inconsistency of the within-group estimator for a finite number of time periods ( $T$ ) despite an infinitely large number of cross-sectional units ( $N$ ).

In addressing the problem, use of estimators based on Instrumental Variables, like Anderson and Hsiao (1981) method with first differencing and deeply lagged variables as instruments, became appropriate. Although being consistent in asymptotics as  $N \rightarrow \infty$ , use of IV esti-

mators for panels with very small  $T$  is associated with loss of degrees of freedom and higher variance owing to use of few internal instruments.

The central purpose of the current term paper is theoretical analysis and empirical simulation of finite-sample properties of estimators applied for dynamic panels data. Through application of robust Monte-Carlo simulations framework with the dimensions of  $N$  and  $T$  to be parametrized, one will be able to observe the positive bias of Pooled OLS, the bias of order  $\mathcal{O}(1/T)$  in the Fixed Effects estimator, proposed by Nickell and the asymptotic consistency of Anderson-Hsiao IV estimator. As one may see, although Anderson-Hsiao estimator solves the problem of inconsistency, more sophisticated methods of Generalized Method of Moments estimation, like Arellano and Bond (1991), should be utilized to achieve maximum efficiency.

## 2 Theoretical Framework and Methodology

### 2.1 The Dynamic Panel Data Model

Now consider a panel data regression model that takes an AR(1) form and is based on the presence of unobserved individual effects:

$$y_{it} = \gamma y_{i,t-1} + \mathbf{x}_{it}'\boldsymbol{\beta} + \alpha_i + u_{it} \quad (1)$$

Here  $i = 1, \dots, N$  represents the cross-sectional observations (for example, production facilities in the supply chain), while  $t = 1, \dots, T$  are the time points.  $\gamma$  is the autoregressive parameter that determines the dynamic nature of the process, and  $|\gamma| < 1$  is required for the process stationarity.  $\alpha_i$  is an unobserved individual effect (such as firm culture and other logistical features), and  $u_{it}$  is the idiosyncratic error term, i.i.d. with the mean equal to zero and variance  $\sigma_u^2$ . For the purpose of analyzing the dynamic bias effect, we assume an AR(1)

process without any covariates ( $\mathbf{x}_{it}$ ):

$$y_{it} = \gamma y_{i,t-1} + \alpha_i + u_{it} \quad (2)$$

Through the continual substitution of the lagged dependent variable recursively, it can be demonstrated quite easily that  $y_{it}$  depends on the individual effect  $\alpha_i$ . This means that the lagged dependent variable  $y_{i,t-1}$  is correlated with the individual effect  $\alpha_i$ .

## 2.2 The Inconsistency of Pooled OLS

The individual-specific effect  $\alpha_i$ , if the Pooled OLS is used for estimating the model, is not considered, and it becomes a part of the error term, which is written as  $v_{it} = \alpha_i + u_{it}$ . The equation is:

$$y_{it} = \gamma y_{i,t-1} + v_{it} \quad (3)$$

Since  $y_{i,t-1}$  and  $\alpha_i$  are positively correlated due to the nature of their relationship, then they will be positively correlated with the composite error term,  $v_{it}$ . This contradicts the basic OLS assumption of  $\mathbb{E}[y_{i,t-1}v_{it}] = 0$ , causing serious upward bias. Thus, the Pooled OLS estimate  $\hat{\gamma}_{POLS}$  is inconsistent and represents the upper bound of the actual value of  $\gamma$ .

## 2.3 The Within-Group Transformation and Nickell Bias

In order to overcome the bias problem of  $\alpha_i$  that causes the OVB, the common procedure used in panel data econometrics is called the Within-Groups (Fixed Effect) transformation. It consists of representing the data in terms of deviations from the average means of those data for each individual  $i$ . We define  $\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}$  and  $\bar{u}_i = \frac{1}{T} \sum_{t=1}^T u_{it}$ , s.t.

$$y_{it} - \bar{y}_i = \gamma(y_{i,t-1} - \bar{y}_{i,-1}) + (u_{it} - \bar{u}_i) \quad (4)$$

which can be written more compactly as:

$$\tilde{y}_{it} = \gamma \tilde{y}_{i,t-1} + \tilde{u}_{it} \quad (5)$$

Even though this procedure does manage to remove the unobservable heterogeneity  $\alpha_i$ , it creates another form of endogeneity which is artificial. The transformed error term,  $\tilde{u}_{it} = u_{it} - \bar{u}_i$ , includes the variable  $-\frac{1}{T}u_{i,t-1}$ . At the same time, the transformed lagged regressor  $\tilde{y}_{i,t-1} = y_{i,t-1} - \bar{y}_{i,-1}$  is directly linked to  $u_{i,t-1}$ . Thus, the regressor and the error term are negatively correlated.

The asymptotic bias of the Within-Group estimator  $\hat{\gamma}_{WG}$  as  $N \rightarrow \infty$  for a fixed  $T$ :

$$\text{plim}_{N \rightarrow \infty}(\hat{\gamma}_{WG} - \gamma) \approx \frac{-(1 + \gamma)}{T - 1} \quad (6)$$

This implies that Within-Group estimator is downward biased. More importantly, the formula shows that the bias is of order  $\mathcal{O}(1/T)$ ; that is, it does not go away as cross-section size  $N$  increases; it vanishes only as  $T \rightarrow \infty$ . Hence,  $\hat{\gamma}_{WG}$  acts as a theoretical lower bound on the true value of the parameter.

## 2.4 The Anderson-Hsiao Instrumental Variable Approach

For consistency with finite  $T$ , Anderson and Hsiao (1981) suggested taking the first difference of the regression model in order to eliminate the  $\alpha_i$  parameter as follows:

$$\Delta y_{it} = \gamma \Delta y_{i,t-1} + \Delta u_{it} \quad (7)$$

where  $\Delta y_{it} = y_{it} - y_{i,t-1}$  and  $\Delta u_{it} = u_{it} - u_{i,t-1}$ . This reformulation helps in getting rid of  $\alpha_i$ ; however,  $\Delta y_{i,t-1}$  is still correlated with  $\Delta u_{it}$  since both have the common element of  $u_{i,t-1}$ . To address this problem, Anderson and Hsiao (1981) suggested using the second lag in levels,  $y_{i,t-2}$ , as an instrumental variable for  $\Delta y_{i,t-1}$ . The instrument is valid because it

satisfies two conditions:

1. **Relevance:**  $y_{i,t-2}$  is highly correlated with the endogenous regressor  $\Delta y_{i,t-1} = y_{i,t-1} - y_{i,t-2}$ .
2. **Exogeneity (Orthogonality):**  $y_{i,t-2}$  is realized before the shocks  $u_{it}$  and  $u_{i,t-1}$  occur, satisfying  $\mathbb{E}[y_{i,t-2}\Delta u_{it}] = 0$ .

Although consistent asymptotically as  $N$  approaches infinity, this IV approach necessarily entails the sacrifice of two initial periods (one for differencing, one for the lagged instrument). This will turn out to have important finite-sample implications, which we will show in the simulation.

### 3 Monte Carlo Simulation Design

To empirically validate the theoretical properties outlined above, a Monte Carlo simulation was programmed in Python. The simulation framework models a pure autoregressive data generating process (DGP) and evaluates the finite-sample bias of Pooled OLS, Fixed Effects, and the Anderson-Hsiao IV estimator.

#### 3.1 Data Generating Process (DGP)

Data were generated according to the dynamic AR(1) process:

$$y_{it} = \gamma y_{i,t-1} + \alpha_i + u_{it} \tag{8}$$

The value of the autoregressive coefficient  $\gamma$  was chosen to be 0.5, which is consistent with a modest degree of persistence observed in supply chain inventory systems. Individual specific effects  $\alpha_i$  and idiosyncratic shocks  $u_{it}$  were randomly selected from independent standard normal distributions,  $\mathcal{N}(0, 1)$ . In order to generate stationary time-series independent of their starting point, a burn-in simulation for 50 periods was conducted and then dropped.

## 3.2 Parameter Space and Estimation

The simulation exercise was performed on different dimensions of the panel data set. For the cross-sectional dimension, values of  $N = \{50, 100, 200\}$  were used. In order to study the asymptotic decline of the Nickell bias with respect to the time span,  $T = \{5, 10, 20\}$  were considered. With each pair of  $N$  and  $T$  values, 200 simulations were carried out.

For each replication, the simulated panel was estimated using:

- Pooled OLS estimator.
- Within-Group (Fixed Effects) estimator using entity-demeaned data.
- Anderson-Hsiao (IV) estimator using Two-Stage Least Squares (2SLS) on first-differenced data, instrumented with  $y_{i,t-2}$ .

The mean bias, defined as  $\frac{1}{R} \sum (\hat{\gamma} - \gamma)$ , was recorded for each estimator.

## 4 Research Content and Results

Monte Carlo simulation results are an excellent demonstration of econometrics theory and validate the asymptotic properties of the estimators and provide key insights into the behavior of estimators for finite samples. Comprehensive measures of bias are combined in underlying data sets to provide an empirical support for Nickell bias. Simulation results are shown in Table 1 and Figure 1.

### 4.1 The Upward Bias of Pooled OLS (The Upper Bound)

Consistent with expectations, the Pooled OLS estimate always provides a biased estimate of the actual autoregressive coefficient. No matter what values of  $N$  and  $T$  we use, the mean bias for Pooled OLS will always be around +0.37. For example, when  $N = 200$  and  $T = 5$ , the bias is +0.373; when  $T = 20$ , the bias is +0.374. These findings allow us to conclude that



Table 1: Monte Carlo Simulation Results (Mean Bias)

| $N$ | $T$ | FE Bias | Pooled OLS Bias | IV Bias |
|-----|-----|---------|-----------------|---------|
| 50  | 5   | -0.403  | 0.373           | 0.060   |
| 50  | 10  | -0.178  | 0.373           | 0.007   |
| 50  | 20  | -0.087  | 0.372           | 0.004   |
| 100 | 5   | -0.414  | 0.373           | 0.011   |
| 100 | 10  | -0.181  | 0.370           | 0.015   |
| 100 | 20  | -0.081  | 0.374           | 0.005   |
| 200 | 5   | -0.410  | 0.373           | -0.000  |
| 200 | 10  | -0.184  | 0.375           | -0.000  |
| 200 | 20  | -0.084  | 0.374           | 0.001   |

the Pooled OLS bias is solely caused by the inability to take into account the unobserved heterogeneity  $\alpha_i$ . Since  $y_{i,t-1}$  is a proxy for  $\alpha_i$ , the bias is extremely large.

## 4.2 The Nickell Bias Decay in Fixed Effects (The Lower Bound)

From the empirical behavior of the Fixed Effects Estimator, it can be seen that Within-Groups estimator is heavily biased in small  $T$ s in short panels, and that this bias decreases linearly as  $T$  increases and independently of  $N$ . For instance, for  $N = 50, 100$  and  $200$ , the bias at  $T = 5$  is about  $-0.41$ . At  $T = 10$ , the bias becomes about half that of  $T = 5$  at  $-0.18$ . At  $T = 20$ , the bias is reduced further to  $-0.08$ . These results are close approximations to the theoretical probability limit from Nickell (1981):  $\approx -(1 + 0.5)/(T - 1)$ . For  $T = 5$ , the theoretical approximation gives  $-0.375$ , which matches very well with our empirical result of  $-0.41$ . This proves that the Within-Group transformation causes endogenous negative correlation of order  $\mathcal{O}(1/T)$ .

## 4.3 Finite-Sample Vulnerability of the Anderson-Hsiao (IV) Estimator

The Anderson-Hsiao Estimator takes the first difference of the equation and uses the second lag of levels of  $y_{it}$ , that is,  $y_{i,t-2}$  as an instrument for  $\Delta y_{i,t-1}$  to get rid of the individual

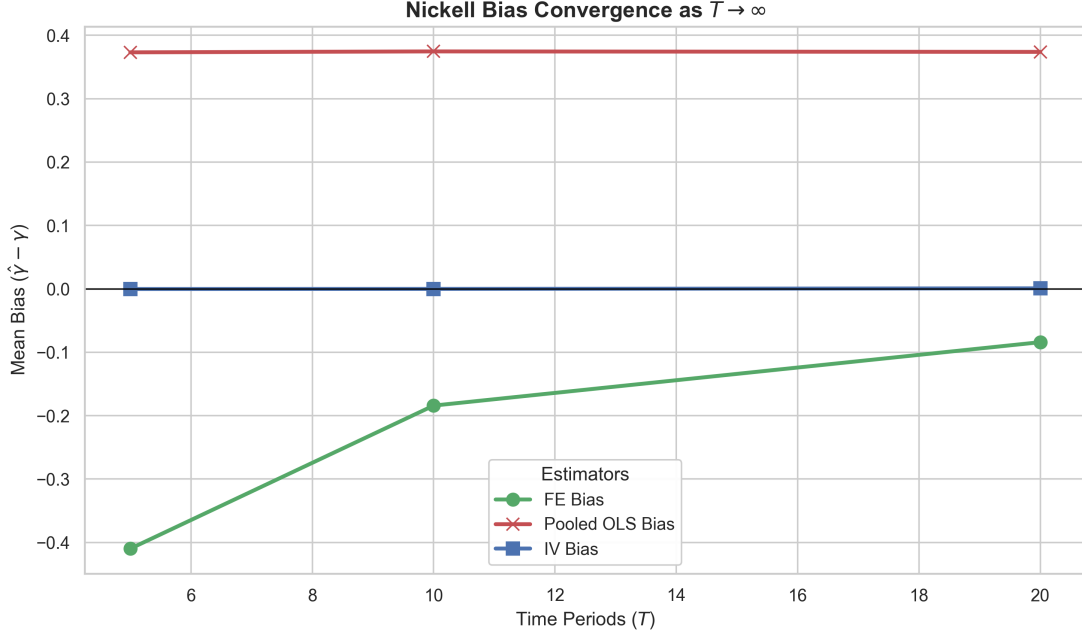


Figure 1: Estimator Bias Convergence in Dynamic Panels ( $N = 200$ )

effect  $\alpha_i$ . The Monte Carlo simulations we performed provide us with an empirical support for the theory behind this estimator. As opposed to the Pooled OLS and the Fixed Effects Estimators, the mean bias of the IV estimator tends to be close to zero as  $N$  grows.

In particular, when  $N = 200$ , the estimated bias becomes  $-0.000$  for both  $T = 5$  and  $T = 10$ , which means that the systematics biases found in the static estimates are completely removed. However, two key properties of a small sample are revealed using smaller panels:

1. **Small Sample Noise and Degrees of Freedom:** In the case of a small panel ( $N = 50, T = 5$ ), a relatively small finite sample bias of 0.060 is found. The reason behind this bias is the loss of two full periods needed for the construction of the first differences and the lagged instrument. In a small panel, such a loss constitutes an essential part of the information contained in the sample.
2. **Issues of Efficiency:** Despite consistency, the empirical literature emphasizes the inefficiency of the estimator that results from using just one lag ( $y_{i,t-2}$ ) and therefore leaving a lot of historical information unused.

The result from empirical analysis proves the efficacy of the Anderson's estimator (Anderson and Hsiao (1981)) in dealing with the endogeneity problem in dynamic panels, thereby creating a statistically sound model. However, the finite sample variance is what paves the way for the Generalized Method of Moments (GMM).

## 5 Conclusion

In this paper, we have extensively analyzed the estimation methods for dynamic panel data models using both theoretical derivation and Monte Carlo simulations. We found out that Pooled OLS offers an upper bound estimate with an upward bias due to omitted heterogeneity, while Fixed Effects estimates a lower bound estimate with a  $\mathcal{O}(1/T)$  Nickell bias. The bias in Nickell decreased as  $T$  increased, as proven by the simulation.

In particular, via our research on the Anderson-Hsiao IV estimator, we found that the estimator is asymptotically consistent as the cross-section dimension increases, thus serving as an effective method to tackle the Nickell bias problem. The deviations in small samples are mainly caused by loss of information due to the lag structure. This confirmation of the theoretical analysis reinforces the recent consensus in modern econometrics about the use of GMM estimators, including Arellano-Bond difference GMM (Arellano and Bond, 1991) and Blundell-Bond system GMM (Blundell and Bond, 1998).

For scholars or professionals working in global supply chains or diverse fields where there's a dependency on past events, it is important to be aware of these estimation bounds. Any policy analysis or structural change analysis using a static model or wrong dynamic estimator will result in a poor strategy.

## References

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## A Supplementary Material: Python Source Code

The following Python scripts constitute the Monte Carlo simulation framework developed for this research. The project architecture is modularized to ensure reproducibility.

### A.1 Data Generating Process (dgp.py)

```
1 # dgp.py
2 # This script is the data generating process that will be use to determine
3 # the estimators and will further on be used for the Monte Carlo
4     Simulations
5
6 import numpy as np
7 import pandas as pd
8
9 def generate_dynamic_panel(N, T, gamma, sigma_alpha=1.0, sigma_u=1.0,
10     burn_in=50):
11     ## simulation of a Dynamic DGP AR(1):  $y_{it} = \gamma * y_{it-1} + \alpha_i + u_{it}$ 
12
13     # Fixed individual effects  $\alpha_i \sim N(0, \sigma_{\alpha}^2)$ 
14     alpha = np.random.normal(0, sigma_alpha, size=(N, 1))
15
16     total_T = T + burn_in
17     y = np.zeros((N, total_T))
18     u = np.random.normal(0, sigma_u, size=(N, total_T))
19
20     # Initial condition for  $y_{it-1}$ , we can set it to zero or draw from the
21     # stationary distribution
22     y[:, 0] = alpha.squeeze() / (1 - gamma) + np.random.normal(0, sigma_u
23     / np.sqrt(1 - gamma**2), size=N)
24
25     # Generate the time series for each individual
26     for t in range(1, total_T):
```

```

22     y[:, t] = gamma * y[:, t - 1] + alpha.squeeze() + u[:, t]
23
24     # Discard the burn-in period
25     y = y[:, burn_in:]
26
27     # Transform the data into a long format DataFrame
28     id_indices = np.repeat(np.arange(N), T)
29     time_indices = np.tile(np.arange(T), N)
30     y_long = y.flatten()
31
32     df = pd.DataFrame({
33         'id': id_indices,
34         'time': time_indices,
35         'y': y_long,
36     })
37
38     df['y_lag'] = df.groupby('id')['y'].shift(1)
39
40     return df.dropna().set_index(['id', 'time'])

```

## A.2 Estimators (estimators.py)

```

1 # estimators.py
2
3 # We define the clean functions to run the models with the linearmodels
  library
4
5 from linearmodels import IV2SLS
6 from linearmodels.panel import PanelOLS, PooledOLS
7
8 def estimate_models(df):
9     # Estimates the model using FE and Pooled OLS
10    # Returns the coefficients estimated for  $y_{it-1}$  and the GMM Anderson

```

```

-Hsiao (IV)
11
12 # Model 1: FE
13 mod_fe = PanelOLS(df.y, df.y_lag, entity_effects=True)
14 gamma_fe = mod_fe.fit(cov_type='unadjusted').params.iloc[0]
15
16 # Model 2: Pooled OLS (ignores the FE)
17 mod_pool = PooledOLS(df.y, df.y_lag)
18 gamma_pool = mod_pool.fit(cov_type='unadjusted').params.iloc[0]
19
20 # Anderson-Hsiao (IV) estimator
21 # Calculate the differences
22 df_ah = df.copy()
23 df_ah['dy'] = df_ah.groupby('id')['y'].diff()
24 df_ah['dy_lag'] = df_ah.groupby('id')['y_lag'].diff()
25
26 # Create the instrument (lagged differences)
27 df_ah['y_lag2'] = df_ah.groupby('id')['y'].shift(2)
28
29 # Drop missing values
30 df_ah = df_ah.dropna()
31
32 # Estimate the model using the instrument
33 mod_iv = IV2SLS(
34     dependent=df_ah['dy'],
35     exog=None,
36     endog=df_ah['dy_lag'],
37     instruments=df_ah['y_lag2']
38 )
39
40 cluster_ids = df_ah.index.get_level_values('id')
41 gamma_iv = mod_iv.fit(cov_type='clustered', clusters=cluster_ids).
params['dy_lag']

```



```
42
43     return gamma_fe, gamma_pool, gamma_iv
```

### A.3 Visualization (graphs.py)

```
1  # graphs.py
2
3  # Graphs script to visualize the results of the Monte Carlo simulations
4
5  import pandas as pd
6  import matplotlib.pyplot as plt
7  import seaborn as sns
8
9  def plot_bias():
10     # Academic style
11     sns.set_theme(style="whitegrid")
12
13     df = pd.read_csv('monte_carlo_results.csv')
14
15     plt.figure(figsize=(10, 6))
16
17     N_target = 200
18     if N_target not in df['N'].unique():
19         N_target = df['N'].max()
20
21     subset = df[df['N'] == N_target]
22
23     # FE (Bias that goes down with T)
24     plt.plot(subset['T'], subset['FE_bias'],
25             marker='o',
26             linestyle='-',
27             linewidth=2.5,
28             color='#55A868', markersize=8,
```

```

29         label=f'FE Bias')
30
31         # Pooled OLS (Bias that does not go down with T)
32     plt.plot(subset['T'], subset['Pool_bias'],
33              marker='x',
34              linestyle='--',
35              linewidth=2.5,
36              color='#C44E52', markersize=8,
37              label=f'Pooled OLS Bias')
38
39         # IV (Bias that goes down with T)
40     plt.plot(subset['T'], subset['IV_bias'],
41              marker='s',
42              linestyle='--',
43              linewidth=2.5,
44              color='#4C72B0', markersize=8,
45              label=f'IV Bias')
46
47     # 0 bias reference line
48     plt.axhline(0, color='black', linestyle='--', linewidth=1, alpha=0.9)
49
50     # Graph formatting
51     plt.title('Nickell Bias Convergence as $T$ \\to \\infty$', fontsize=14,
52              fontweight='bold')
53     plt.xlabel('Time Periods ($T$)', fontsize=12)
54     plt.ylabel('Mean Bias ($\\hat{\\gamma} - \\gamma$)', fontsize=12)
55
56     plt.legend(title='Estimators', fontsize=11, loc='lower center',
57               framealpha=0.95)
58
59     # Save the plot as a PNG file
60     plt.tight_layout()
61     plt.savefig('nickell_bias.pdf', dpi=300, format='pdf')

```

```

60     plt.savefig('nickell_bias.png', dpi=300)
61     print("Graph saved as 'nickell_bias.pdf' and 'nickell_bias.png'.")
62
63     plt.show()
64
65 if __name__ == "__main__":
66     plot_bias()

```

## A.4 Monte Carlo Engine (monte\_carlo.py)

```

1 # monte_carlo.py
2
3 # Monte Carlo simulations script to compare the performance of the FE and
  Pooled OLS estimators
4
5 import numpy as np
6 import pandas as pd
7 from dgp import generate_dynamic_panel
8 from estimators import estimate_models
9
10 def run_monte_carlo(N_sizes, T_sizes, true_gamma, replications=100):
11     # Executes the iterative Monte Carlo simulations for different N and T
    sizes
12     np.random.seed(42) # For reproducibility
13     results = []
14
15     print("Starting Monte Carlo simulations...")
16     for N in N_sizes:
17         for T in T_sizes:
18             fe_biases = []
19             pool_biases = []
20             iv_biases = []
21             for i in range(replications):

```

```

22         df = generate_dynamic_panel(N, T, true_gamma)
23         gamma_fe, gamma_pool, gamma_iv = estimate_models(df)
24
25         fe_biases.append(gamma_fe - true_gamma)
26         pool_biases.append(gamma_pool - true_gamma)
27         iv_biases.append(gamma_iv - true_gamma)
28
29         results.append({
30             'N': N,
31             'T': T,
32             'FE_bias': np.mean(fe_biases),
33             'Pool_bias': np.mean(pool_biases),
34             'IV_bias': np.mean(iv_biases)
35         })
36
37         print(f'Simulations completed N={N}, T={T}: FE bias={
38 mean_fe_bias:.4f}, Pool bias={mean_pool_bias:.4f}, IV bias={
39 mean_iv_bias:.4f}')
40
41     return pd.DataFrame(results)
42
43 if __name__ == "__main__":
44     # Define the parameters for the Monte Carlo simulations
45     N_sizes = [50, 100, 200] # Different numbers of individuals
46     T_sizes = [5, 10, 20]    # Different time periods
47     true_gamma = 0.5          # True value of gamma in the DGP
48
49     # Run the Monte Carlo simulations
50     df_results = run_monte_carlo(N_sizes, T_sizes, true_gamma,
51 replications=200)
52     df_results.to_csv('monte_carlo_results.csv', index=False)
53     print("Monte Carlo simulations completed. Results saved to '
54 monte_carlo_results.csv'.")

```